

Executive Summary

Total draws analysed: 3116

Date range: 19/11/1994 to 01/11/2025

Methodology — Clustering & Model Selection

K-Means clustering is applied to the six main drawn numbers for each draw to explore potential groupings. The primary quantitative selection metric is the mean silhouette score evaluated for $K = 2..10$. Because lottery draws are high-noise and near-random, absolute silhouette values are expected to be low; we therefore use per-K diagnostic visualisations (per-sample silhouette distributions, cluster size balance, centroid dispersion) and PCA projections for interpretability. The selected K is chosen to maximise silhouette while also ensuring interpretable and reasonably balanced clusters. PCA is solely used for visualization and does not affect model fitting.

Ball Frequency Overview

Most Frequent Balls: 39, 38, 11, 33, 27, 31

Middle Frequent Balls: 36, 28, 45, 18, 48, 8

Least Frequent Balls: 57, 51, 56, 50, 53, 55

1. Recent 20 Draws (Newest First, with Cluster)

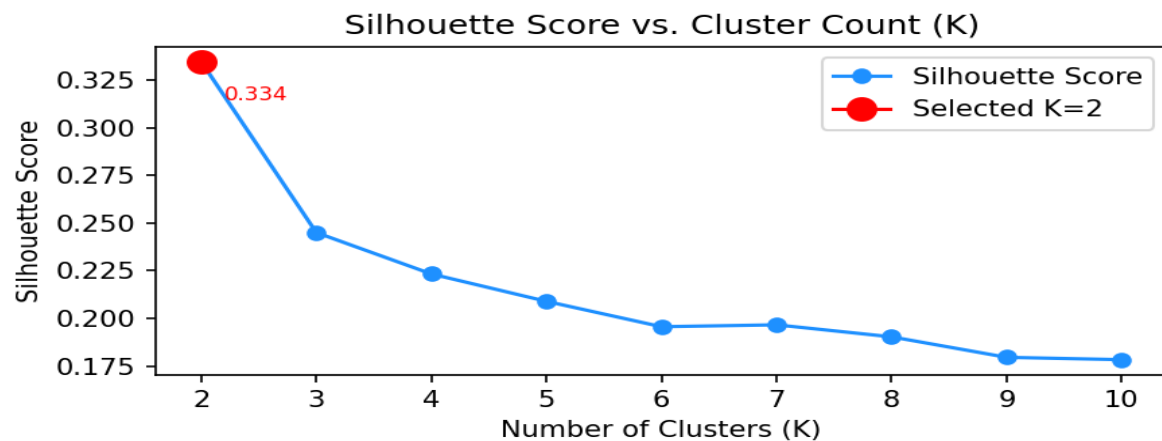
DrawDate	Ball1	Ball2	Ball3	Ball4	Ball5	Ball6	BN	Jackpot	Wins	Machine	Set	Cluster
01/11/2025	17	18	31	36	39	51	29	3800000.0	0.0	nan	nan	1.0
29/10/2025	9	24	30	33	35	54	32	12423596.0	0.0	nan	nan	1.0
25/10/2025	4	9	28	40	42	59	55	10513580.0	0.0	nan	nan	1.0
22/10/2025	3	24	26	34	35	36	9	8360535.0	0.0	nan	nan	1.0
18/10/2025	6	39	43	52	58	59	19	7053955.0	0.0	nan	nan	1.0
15/10/2025	29	33	36	38	48	58	52	5057815.0	0.0	nan	nan	1.0
11/10/2025	11	17	25	31	51	54	53	3800000.0	0.0	nan	nan	1.0
08/10/2025	2	21	42	48	56	57	51	2000000.0	1.0	nan	nan	1.0
04/10/2025	6	8	12	33	49	59	42	10633323.0	1.0	nan	nan	0.0
01/10/2025	13	28	31	39	51	59	27	8458823.0	0.0	nan	nan	1.0
27/09/2025	15	23	24	28	37	41	59	7075979.0	0.0	nan	nan	1.0
24/09/2025	3	16	23	30	38	47	25	5029330.0	0.0	nan	nan	0.0
20/09/2025	13	16	30	36	53	59	22	3800000.0	0.0	nan	nan	1.0
17/09/2025	7	9	18	24	30	36	42	1000000.0	4.0	nan	nan	0.0
13/09/2025	1	26	36	49	53	54	10	15000000.0	0.0	nan	nan	1.0
10/09/2025	1	9	18	27	38	42	13	8430747.0	0.0	nan	nan	0.0
06/09/2025	10	22	37	40	45	57	24	7082906.0	0.0	nan	nan	1.0
03/09/2025	7	15	18	22	31	54	28	5063522.0	0.0	nan	nan	0.0
30/08/2025	6	10	19	20	43	59	16	3800000.0	0.0	nan	nan	0.0
27/08/2025	5	25	43	45	47	55	58	8655122.0	1.0	nan	nan	1.0

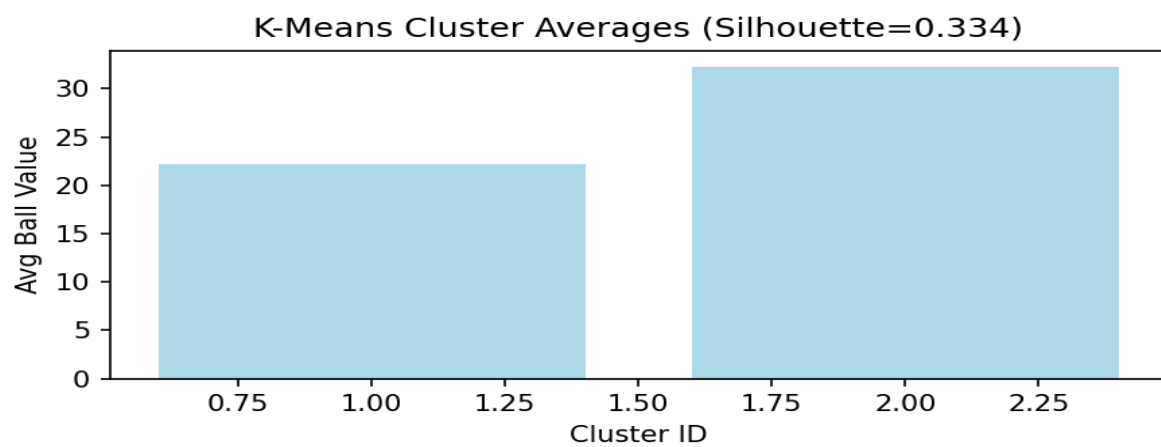
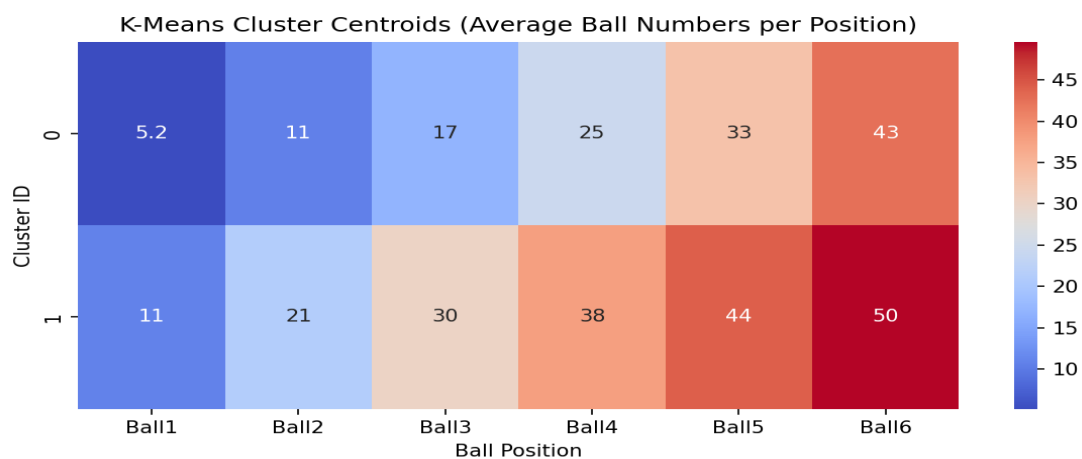
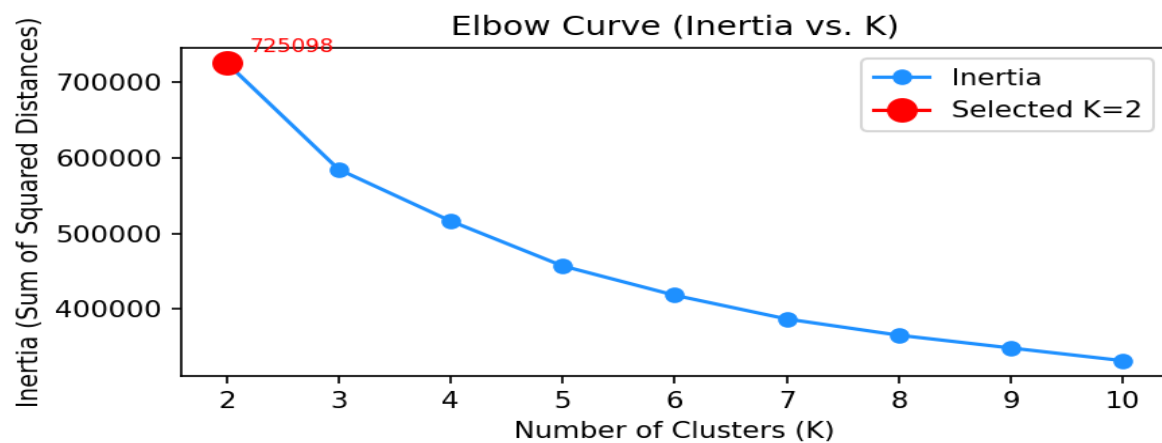
2. Ball Reoccurrence Analysis

Category	Ball	Frequency
Most Frequent	39	392
Most Frequent	38	390
Most Frequent	11	389
Most Frequent	33	387
Most Frequent	27	385
Most Frequent	31	383
Middle	36	357
Middle	28	357
Middle	45	356
Middle	18	356
Middle	48	355
Middle	8	355
Least Frequent	57	106
Least Frequent	51	104
Least Frequent	56	102
Least Frequent	50	100
Least Frequent	53	100
Least Frequent	55	98

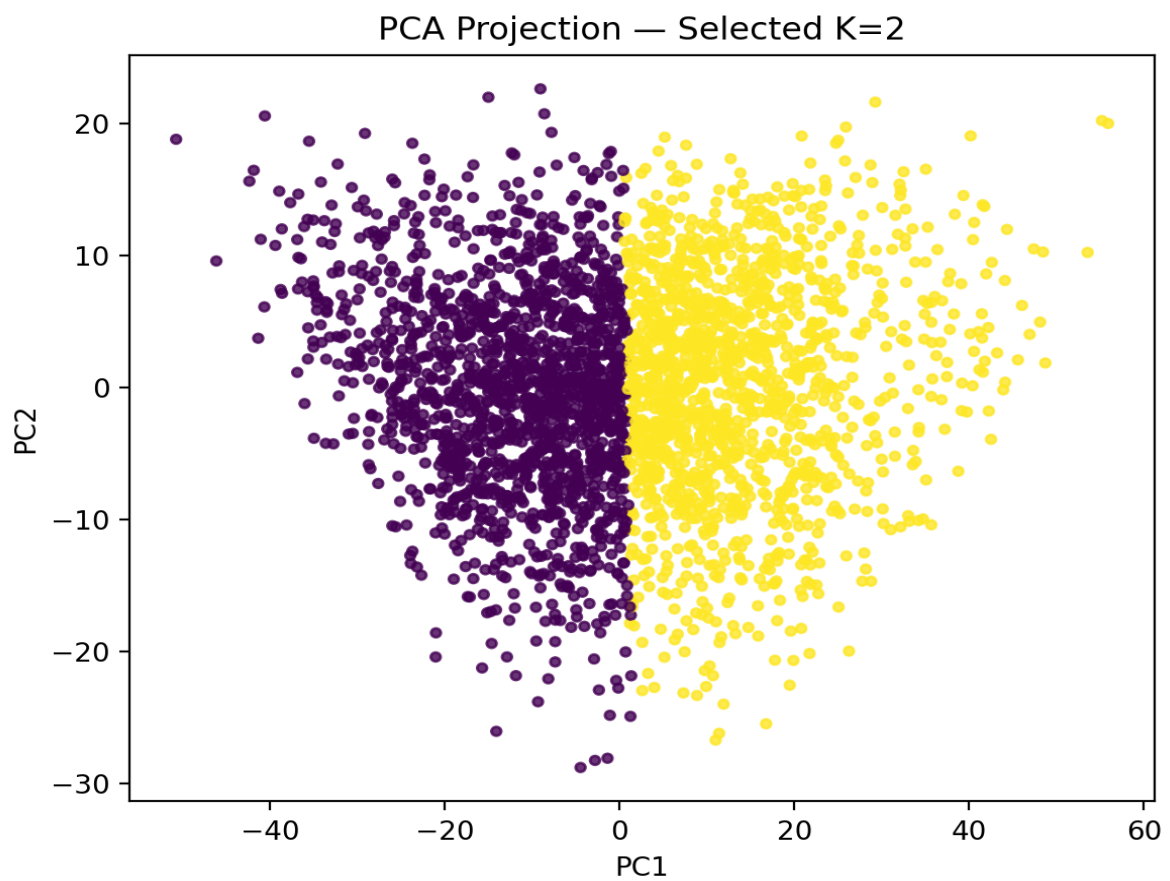
3. K-Means Clustering Analysis

K-Means performed with k=2 clusters. Silhouette Score = 0.334.





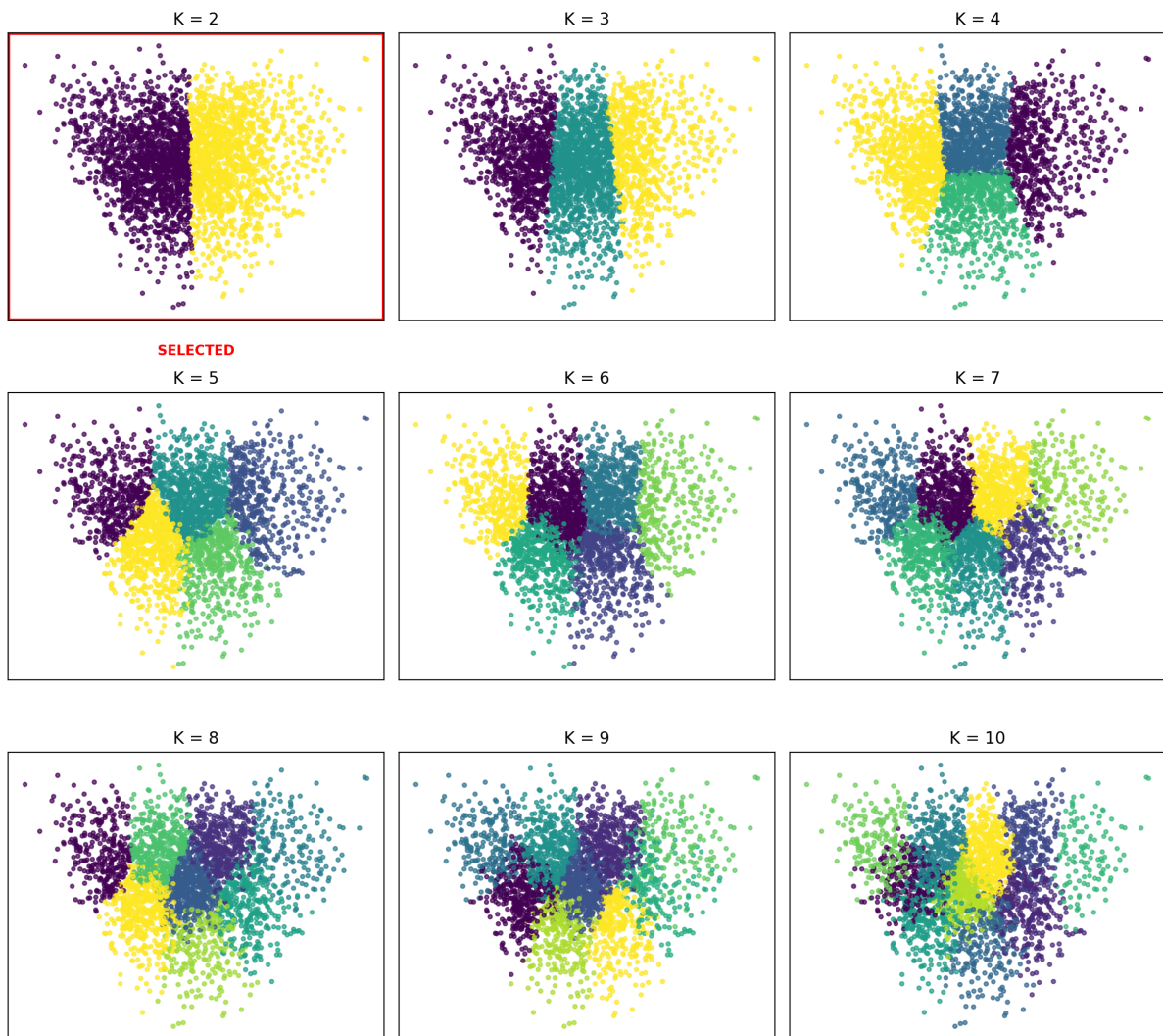
PCA Projection — Selected K



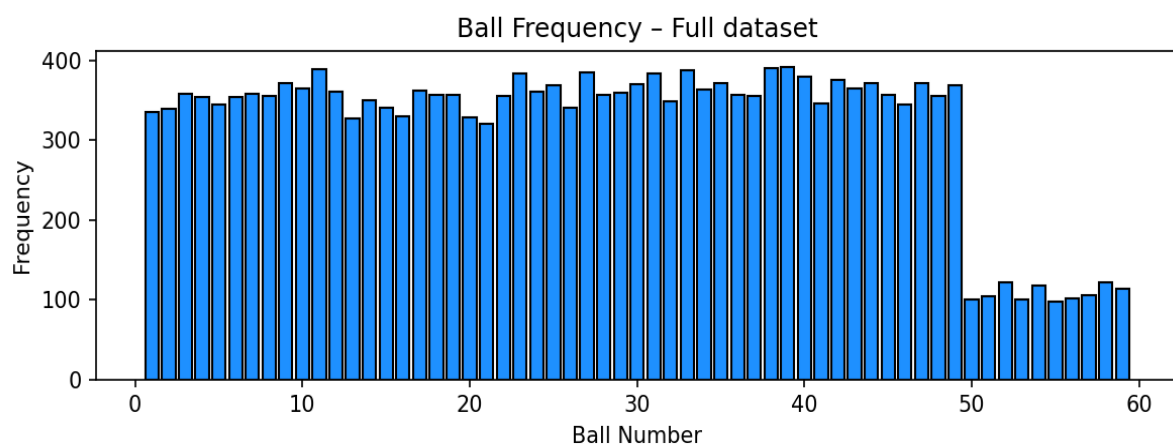
PCA Grid — Comparison Across K (2–10)

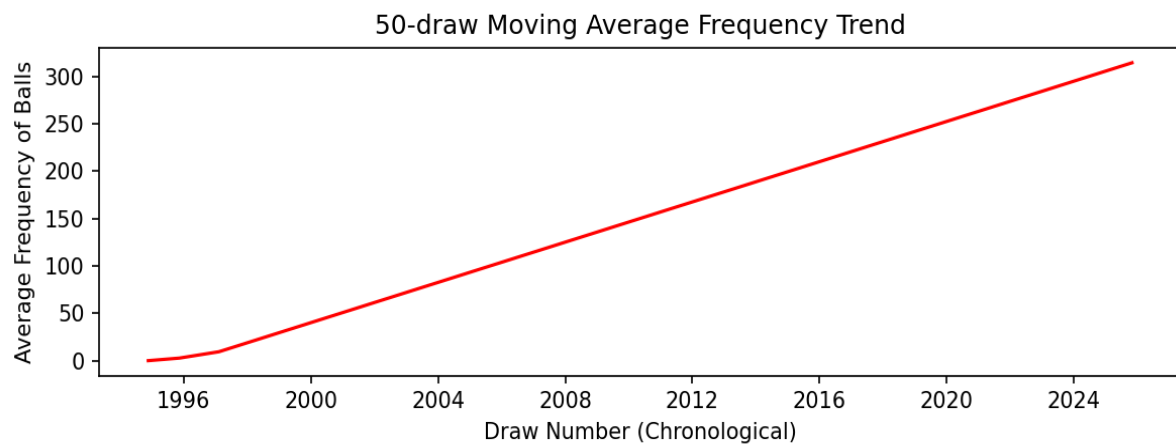
Selected K is highlighted in red on the grid.

PCA Projection of Lottery Draws (K = 2-10)



4. Frequency Distribution





5. Conclusions

- Optimal cluster count: 2 (Silhouette=0.334).
- Ball frequency distribution shows mild decade bias.
- No repeating 6-number lines observed in full dataset.